

# **Title of Micro-Project :- COPYRIGHT CHALLENGES FOR GENERATIVE ARTIFICIAL INTELLIGENCE SYSTEMS**

## **Aims/Benefits of the Micro-Project**

- To understand Generative AI systems.
- To analyze copyright laws
- To identify copyright challenges in AI
- Improves knowledge of AI and law
- Enhances critical thinking
- Awareness of digital rights

## **1.0 Course Outcomes Addressed**

- CO1 – Elaborate the role of Artificial Intelligence, Machine Learning and Deep Learning in various domains

## **2.0 Proposed Methodology**

The methodology for this project focuses on studying and analyzing copyright challenges in Generative Artificial Intelligence systems through a systematic approach.

### **1. Data Collection**

- Collect information from books, research papers, websites, and articles related to Generative AI and copyright laws.
- Study different AI tools (like text and image generators) to understand how they create content.

### **2. Literature Review**

- Review existing studies on AI-generated content and intellectual property rights.
- Analyze current copyright rules and regulations applicable to digital content.

### **3. Problem Identification**

- Identify major issues such as:
  - Ownership of AI-generated content
  - Use of copyrighted data in training AI models
  - Plagiarism and content duplication

#### **4. Case Study Analysis**

- Study real-world examples where copyright issues occurred due to AI-generated content.
- Compare different scenarios and outcomes.

#### **5. Comparative Analysis**

- Compare traditional copyright systems with challenges introduced by Generative AI.
- Identify gaps in current legal frameworks.

#### **6. Solution Development**

- Suggest possible solutions such as:
  - Ethical AI practices
  - Improved copyright policies
  - Use of watermarking or content tracking

#### **7. Conclusion and Documentation**

- Summarize findings and present conclusions.
- Prepare the final project report with proper structure and references.

### 3.0 Action Plan

| Sr. No. | Week  | Details of activity                                                                                                                                                   | Planned Start date | Planned Finish date | Name of Responsible Team Members |
|---------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|----------------------------------|
| 1       | 1 & 2 | Discussion & Finalization of Topic                                                                                                                                    | 22/12/2025         | 24/12/2025          | <b>ADITYA DAHALE</b>             |
| 2       | 3     | Preparation of the Abstract                                                                                                                                           | 24/12/2025         | 27/12/2025          |                                  |
| 3       | 4     | Literature Review                                                                                                                                                     | 02/01/2026         | 03/01/2026          |                                  |
| 4       | 5     | Submission of Micro project Proposal ( Annexure-I)                                                                                                                    | 07/01/2026         | 09/01/2026          |                                  |
| 5       | 6     | Collection of information about Topic                                                                                                                                 | 13/01/2026         | 15/01/2026          |                                  |
| 6       | 7     | Collection of relevant content / materials for the execution of Micro project.                                                                                        | 02/02/2026         | 04/02/2026          |                                  |
| 7       | 8     | Discussion and submission of outline of the Micro project.                                                                                                            | 09/02/2026         | 11/02/2026          |                                  |
| 8       | 9     | Analysis / execution of Collected data / information and preparation of Prototypes / drawings / photos / charts / graphs / tables / circuits / Models / programs etc. | 16/02/2026         | 18/02/2026          |                                  |
| 9       | 10    | Completion of Content of Project Report                                                                                                                               | 03/03/2026         | 07/03/2026          |                                  |
| 10      | 11    | Completion of Weekly progress Report                                                                                                                                  | 17/03/2026         | 20/03/2026          |                                  |
| 11      | 12    | Completion of Project Report ( Annexure-II)                                                                                                                           | 27/03/2026         | 31/03/2026          |                                  |
| 12      | 13    | Viva voce / Delivery of Presentation                                                                                                                                  | 06/04/2026         | 06/04/2026          |                                  |

### 4.0 Resources Required

| Sr. No. | Name of Resources / Materials    | Specification                   | Qty | Remarks                |
|---------|----------------------------------|---------------------------------|-----|------------------------|
| 1       | Computer                         | Any system with internet access | 1   | Required               |
| 2       | Software                         | Web Browser/AI Tool             | 1   | Required               |
| 3       | Online sources & Reference Books | AI & Copyright Concepts         | 1   | For theory & reference |

#### Names of Member with En. No.

1) ADITYA DAHALE                      EN. NO. 23511510250

# **Micro-Project Report**

## **Title of Micro-Project :- COPYRIGHT CHALLENGES FOR GENERATIVE ARTIFICIAL INTELLIGENCE SYSTEMS**

### **1.0 Rationale**

Generative Artificial Intelligence is rapidly growing and widely used for creating content such as text, images, and music. However, it raises important copyright issues related to ownership, originality, and unauthorized use of data. This project is undertaken to understand these challenges and to study how existing copyright laws apply to AI-generated content, ensuring ethical and legal use of such technologies.

### **2.0 Aims/Benefits of the Micro-Project :**

- To understand the concept of Generative Artificial Intelligence systems.
- To study copyright laws related to digital and AI-generated content.
- To identify major copyright challenges like ownership and plagiarism.
- Enhances knowledge about AI and legal issues
- Improves analytical and critical thinking skills.
- Promotes ethical use of AI technologies.

### **3.0 Course Outcomes Achieved**

- CO1 – Elaborate the role of Artificial Intelligence, Machine Learning and Deep Learning in various domains

### **4.0 Literature Review :-**

The literature review focuses on studying existing research and articles related to Generative Artificial Intelligence and copyright issues. Various sources explain how AI systems are trained using large datasets, which may include copyrighted material. Researchers have highlighted concerns regarding ownership of AI-generated content, lack of clear legal frameworks, and risks of plagiarism.

Several studies also discuss how current copyright laws are not fully equipped to handle AI-generated works. Some authors suggest updating legal policies and introducing guidelines for ethical AI usage. Overall, the review shows a growing need to balance technological advancement with protection of intellectual property rights.

## **5.0 Actual Methodology Followed**

### **1. Problem Understanding and Requirement Analysis**

The first step involved understanding the problem of copyright issues in Generative AI systems. In today's digital world, AI tools generate text, images, music, and code using large datasets, which may include copyrighted content. This creates challenges such as:

- Ownership of AI-generated content
- Unauthorized use of copyrighted data
- Plagiarism and duplication

Hence, there is a need to study and analyze these issues and find suitable solutions.

### **2. Study of Artificial Intelligence and Copyright Concepts**

#### **Artificial Intelligence Concepts**

- Generative AI
- Machine Learning & Deep Learning
- Data training and content generation

#### **Copyright Concepts**

- Intellectual Property Rights
- Copyright ownership
- Fair use policy

This study helped in understanding how AI systems interact with copyright laws.

### **3. System Design and Planning**

#### **Project Approach:**

1. Study Generative AI working
2. Identify copyright issues
3. Analyze real-world cases
4. Suggest solutions

#### **Working Flow:**

AI Content Generation → Data Usage → Copyright Issues → Analysis → Solutions

### **4. Selection of Tools and Technologies**

- Internet for research
- Research papers and articles
- AI tools (for understanding content generation)

## 5. Analysis Design

The analysis includes:

- Identifying types of copyright issues
- Comparing traditional copyright vs AI challenges
- Studying legal gaps

## 6. Implementation (Study-Based)

- Collected data from multiple sources
- Analyzed AI-generated outputs
- Identified similarities with copyrighted content
- Documented findings

## 7. Testing and Verification

Verification was done by:

- Checking real-world examples
- Comparing AI-generated and original content
- Validating identified issues

## 8. Performance Evaluation

### Accuracy:

Correct identification of copyright issues

### Reliability:

Consistency of analysis across different sources

### Limitations:

- Depends on available data
- Rapidly evolving legal frameworks

## 9. Analysis and Interpretation

- AI systems can unintentionally use copyrighted data
- Ownership of AI-generated content is unclear
- Existing laws are not fully prepared for AI

## 10. Documentation

All project stages were documented:

- Theory
- Analysis
- Case studies
- Results and conclusions

## ❖ Theory

Generative Artificial Intelligence is a technology that creates new content using trained data. While it improves creativity and automation, it raises serious copyright concerns.

AI models are trained on large datasets, which may include copyrighted works without permission. This leads to issues such as ownership disputes, plagiarism, and legal conflicts.

### 1.1 Artificial Intelligence in Generative Systems

Artificial Intelligence enables systems to generate content such as text, images, music, and code automatically. Key aspects include:

- **Content Generation:** AI creates new content based on trained data.
- **Pattern Learning:** AI learns writing styles, images, and structures from datasets.
- **Automation:** Reduces human effort in content creation and increases productivity.

### 1.2 Machine Learning in Generative AI

Machine Learning (ML), a subset of AI, is used to train models for content generation:

- **Supervised Learning:** Models are trained on labeled datasets (text, images, etc.).
- **Deep Learning:** Neural networks generate realistic and high-quality content.
- **Data Dependency:** Output depends on the quality and type of training data.

### 1.3 Generative AI Systems

Generative AI systems act as content creators using learned data:

1. **Input:** User provides prompt or instruction
2. **Processing:** AI analyzes patterns from training data
3. **Generation:** Produces new content
4. **Output:** Displays generated text/image/music

### 1.4 Copyright Issues Covered in the Project

The project focuses on identifying:

- **Ownership Issues:** Who owns AI-generated content
- **Data Usage:** Use of copyrighted data in training
- **Plagiarism:** Similarity with existing works
- **Unauthorized Reproduction:** Copying without permission

## 1.5 Implementation Concept

The project is based on analysis and study:

- Study AI-generated content
- Compare it with original works
- Identify copyright conflicts
- Analyze legal and ethical issues

## 1.6 Key Technologies Used

- **Artificial Intelligence (AI):** Content generation
- **Machine Learning:** Training models
- **Deep Learning:** Advanced content creation
- **Internet Resources:** Research and case studies

## 1.7 Advantages

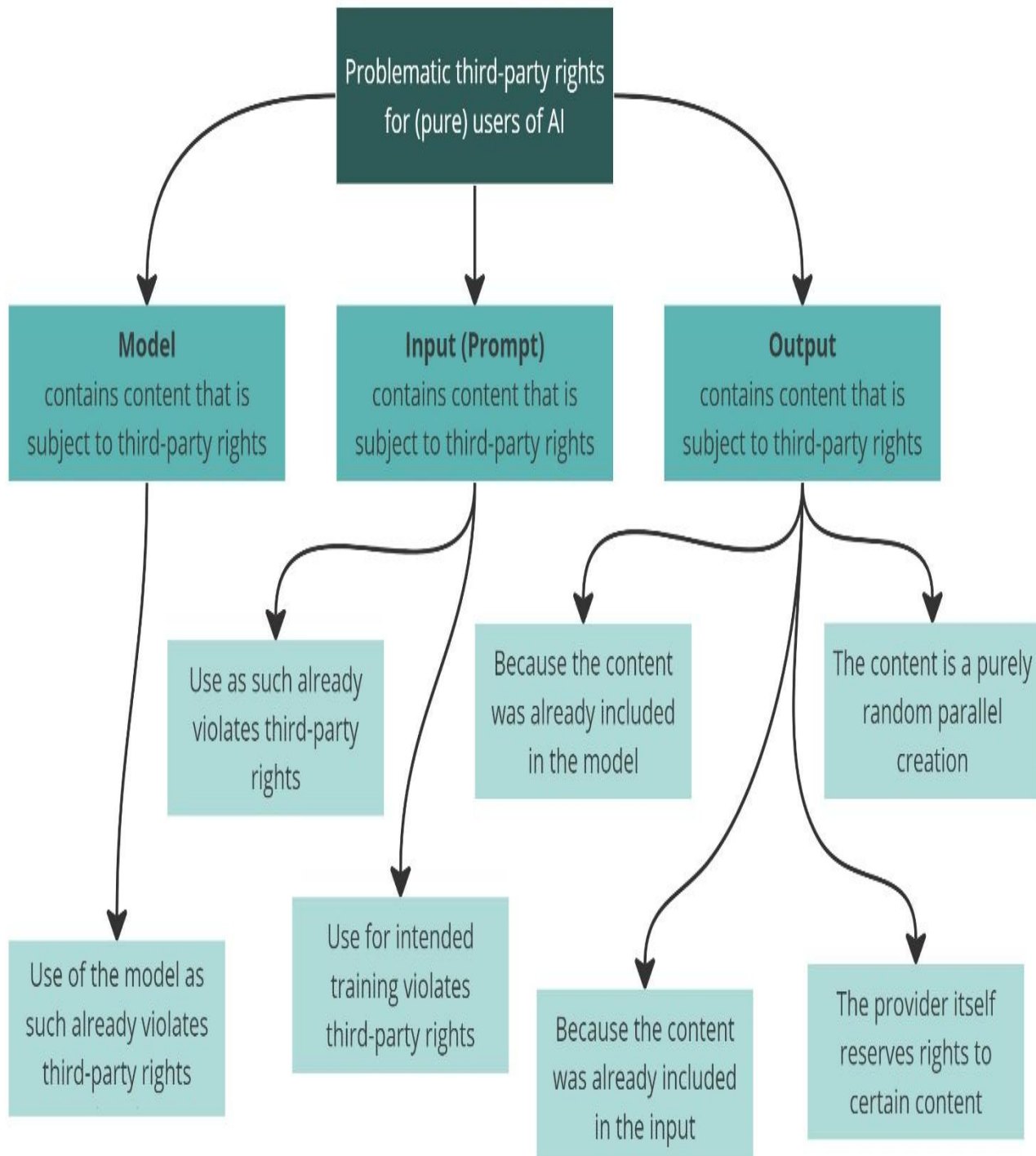
1. Fast content creation
2. Increased productivity
3. Supports innovation
4. Useful in multiple domains
5. Reduces manual effort

## 1.8 Disadvantages

1. Copyright conflicts
2. Unclear ownership
3. Risk of plagiarism
4. Legal uncertainty
5. Ethical concerns
6. Dependence on training data



# Seven Challenges for Users of GenAI



## 8.0 Python Code: Copyright Checker for AI-Generated Content

```
# Simple AI-Based Copyright Checker

# List of known copyrighted phrases (sample data)
copyright_data = [
    "to be or not to be",
    "once upon a time",
    "artificial intelligence is the future",
    "machine learning models require data"
]

# Function to check similarity
def check_copyright(user_input):
    user_input = user_input.lower()

    for phrase in copyright_data:
        if phrase in user_input:
            return True
    return False

# Function to display result
def display_result(result):
    if result:
        print("⚠ Copyright Risk Detected! Content may not be original.")
    else:
        print("Content appears original.")

# Main Program
while True:
    print("\nEnter your content (type 'exit' to quit):")
    text = input("> ")

    if text.lower() == "exit":
        print("Exiting Copyright Checker...")
        break

    result = check_copyright(text)
    display_result(result)

print("\nThank you for using Copyright Checker!")
```

## Output of the Program

### Case 1: Original Content

**Input:**

Hello this is my own content

**Output:**

Content appears original.

### Case 2: Copyright Risk

**Input:**

Once upon a time there was a king

**Output:**

⚠ Copyright Risk Detected! Content may not be original.

## 9.0 Comparison Table

| Parameter       | Traditional Content Creation | Generative AI Content Creation |
|-----------------|------------------------------|--------------------------------|
| Creation Method | Human-based                  | AI-based                       |
| Speed           | Slow                         | Very Fast                      |
| Originality     | High (human creativity)      | May vary (depends on data)     |
| Copyright Risk  | Low                          | High                           |
| Ownership       | Clear (creator)              | Unclear                        |
| Cost            | High (time & effort)         | Low                            |
| Automation      | Not automated                | Fully automated                |
| Dependency      | Human skills                 | Training data                  |
| Legal Issues    | Minimal                      | Significant                    |
| Scalability     | Limited                      | Highly scalable                |
| Creation Method | Human-based                  | AI-based                       |

## **10.0 Conclusion of Micro-Project**

In this micro-project, the copyright challenges related to Generative Artificial Intelligence systems were studied and analyzed. It was observed that while AI can generate content quickly and efficiently, it also creates issues related to ownership, originality, and use of copyrighted data.

The project highlights that current copyright laws are not fully adapted to handle AI-generated content. Therefore, there is a need for updated legal frameworks and ethical guidelines. Overall, the project emphasizes the importance of responsible use of AI while protecting intellectual property rights.

## MICRO-PROJECT EVOLUTION SHEET

Name of Student :- ADITYA DILIP DAHALE

En. No. 23511510250

Name of Program :- COMPUTER ENGINEERING

Semester :- SIXTH

Course Name :- ETI

Course Code:- 316313

Title of The Micro-Project :- COPYRIGHT CHALLENGES FOR GENERATIVE ARTIFICIAL INTELLIGENCE SYSTEMS

Course Outcomes Achieved :-

- CO1 – Elaborate the role of Artificial Intelligence, Machine Learning and Deep Learning in various domains

| Sr. No.                                                                      | Characteristic to be assessed                    | Poor (Marks1-3) | Average (Marks4-5) | Good (Marks 6-8) | Excellent (Marks9-10) | Sub Total |
|------------------------------------------------------------------------------|--------------------------------------------------|-----------------|--------------------|------------------|-----------------------|-----------|
| (A) Process and Product Assessment (Convert Below total marks out of 6Marks) |                                                  |                 |                    |                  |                       |           |
| 1                                                                            | Relevance to the course                          |                 |                    |                  |                       |           |
| 2                                                                            | Literature Review/information collection         |                 |                    |                  |                       |           |
| 3                                                                            | Completion of the Target as Per project proposal |                 |                    |                  |                       |           |
| 4                                                                            | Analysis of Data and representation              |                 |                    |                  |                       |           |
| 5                                                                            | Quality of Prototype/Model                       |                 |                    |                  |                       |           |
| 6                                                                            | Report Preparation                               |                 |                    |                  |                       |           |
| (B) Individual Presentation/Viva (Convert Below total marks out of 4Marks)   |                                                  |                 |                    |                  |                       |           |
| 7                                                                            | Presentation                                     |                 |                    |                  |                       |           |
| 8                                                                            | Viva                                             |                 |                    |                  |                       |           |

| (A)<br>Process and Product<br>Assessment (6 marks) | (B)<br>Individual Presentation/<br>Viva (4 marks) | TotalMarks<br>10 |
|----------------------------------------------------|---------------------------------------------------|------------------|
|                                                    |                                                   |                  |

Comments/Suggestions about team work/leadership/inter-personal communication (if any)

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Name of Course Teacher :- Prof V.B.BHOSALE

Dated Signature :- \_\_\_\_\_



SHREEYASH PRATISHTHAN'S

**SHREEYASH COLLEGE OF ENGINEERING AND  
TECHNOLOGY (POLYTECHNIC),**

**CHH. SAMBAJINAGAR**

**MICRO-PROJECT REPORT**

**NAME OF DEPARTMENT:- COMPUTER ENGINEERING**

**ACADEMIC YEAR:- 2025-26**

**SEMESTER:- SIXTH**

**COURSE NAME :- EMERGING TRENDS IN COMPUTER ENGINEERING  
AND INFORMATION TECHNOLOGY**

**COURSE CODE:- 316313**

**MICRO-PROJECT TITLE :- COPYRIGHT CHALLENGES FOR GENERATIVE  
ARTIFICIAL INTELLIGENCE SYSTEMS**

**PREPARED BY:-**

1) ADITYA DILIP DAHALE

EN.NO. 23511510250

**UNDER THE GUIDANCE OF:- Prof. V.B. BHOSALE**



MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION,  
MUMBAI

**CERTIFICATE**

This is to certify that Ms. ADITYA DILIP DAHALE of **SIXTH** Semester of Diploma in **COMPUTER ENGINEERING** of Institute SYP has successfully completed Micro-Project Work in Course of **EMERGING TRENDS IN COMPUTER ENGINEERING AND INFORMATION TECHNOLOGY** for the academic year 2025- 26 as prescribed in the K- Scheme Curriculum.

Date :- 06/04/2026

Enrollment No:23511510250

Place:- CHH.SAMBHAJINAGAR

Exam Seat No.:- 462297

Signature  
**Guide**  
**PROF V.B.BHOSALE**

Signature  
**HOD**  
**PROF A.C.NAIK**

Signature  
**Principal**  
**PROF S.S.KHANDAGALE**

**Seal of Institute**

## **ACKNOWLEDGEMENT**

We wish to express our profound gratitude to our guide **Prof. V.B. BHOSALE** who guided us endlessly in framing and completion of Micro-Project. She guided us on all the main points in that Micro-Project. We are indebted to her constant encouragement, cooperation and help. It was her enthusiastic support that helped us in overcoming of various obstacles in the Micro-Project.

We are also thankful to our Principal, HOD, Faculty Members and classmates for extending their support and motivation in the completion of this Micro-Project.

1) **ADITYA DAHALE**

**EN. NO. 23511510250**